

# Chapter 4, Section 3

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1. If  $X$  is a curve of genus 2, show that a divisor  $D$  is very ample  $\iff \deg D \geq 5$ . This strengthens (3.3.4).

*Proof.* Let  $D$  be a very ample divisor on  $X$ . By the degree-genus formula, a nonsingular plane curve cannot have genus equal to 2, which implies  $\dim |D| \geq 3$ . Applying the Riemann-Roch theorem to  $D$  gives

$$\deg D = \dim |D| - \dim |K - D| + 1 \geq 4 - \dim |K - D|.$$

By (Ex. 1.5),  $D$  has degree at least 3 (Ex. 1.5), so  $K - D$  has negative degree. Hence,  $|K - D|$  has dimension -1.  $\square$

3. If  $X$  is a curve of genus  $\geq 2$  which is a complete intersection (II, Ex. 8.4) in some  $\mathbb{P}^n$ , show that the canonical divisor  $K$  is very ample. Conclude that a curve of genus 2 can never be a complete intersection in any  $\mathbb{P}^n$ .

*Proof.* Let  $d$  be the degree of  $X$  in  $\mathbb{P}^n$ , and assume  $X$  is not contained in any hyperplane. Then  $X$  is a complete intersection of  $n - 1$  hypersurfaces of degree  $d_i$ , and by (II, Ex. 8.4), we have  $\mathcal{O}_X(K) \cong \mathcal{O}_X(e)$ , where  $e = \sum d_i - n - 1$ , so it is enough to show  $e \geq 1$ . We have two ways of computing  $\deg K$ , which are  $\deg K = ed$  and  $\deg K = 2g - 2 \geq 2$ . Hence,  $e \geq 1$ . The linear system given by the canonical sheaf of a curve of genus 2 is the unique  $g_2^1$  (Ex. 1.7), which is never ample.  $\square$

## 6. Curves of Degree 4.

- (a) If  $X$  is a curve of degree 4 in some  $\mathbb{P}^n$ , show that either

- (1)  $g = 0$ , in which case  $X$  is either the rational normal quartic in  $\mathbb{P}^4$  (Ex. 3.4) or the rational quartic curve in  $\mathbb{P}^3$  (II, 7.8.6), or
- (2)  $X \subseteq \mathbb{P}^2$ , in which case  $g = 3$ , or
- (3)  $X \subseteq \mathbb{P}^3$  and  $g = 1$ .

- (b) In the case  $g = 1$ , show that  $X$  is a complete intersection of two irreducible quadric surfaces in  $\mathbb{P}^3$  (I, Ex. 5.1).

*Proof.*

- (a)  $\boxed{n = 2}$  The degree-genus formula gives

$$g = \frac{(d-1)(d-2)}{2} = 3.$$

$\boxed{n = 3}$  If  $X$  is not contained in any hyperplane, then  $g < 3$  (III, Ex. 9.11) (Ex. 3.5b). Since  $g \neq 2$  by (Ex. 3.1),  $X$  is either the rational quartic curve in  $\mathbb{P}^3$ , or an elliptic curve.

$\boxed{n = 4}$   $X$  is rational by (Ex. 1.5).

- (b) Let  $\mathcal{I}$  be the ideal sheaf of  $X$  in  $\mathbb{P}^3$ , and consider the exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by  $\mathcal{O}_{\mathbb{P}^3}(2)$  and taking cohomology gives

$$0 \longrightarrow H^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow H^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Now,  $X$  is an elliptic curve, and  $\mathcal{O}_X(2)$  corresponds to a divisor  $D$  of degree 8. The Riemann Roch theorem gives  $\dim H^0(X, \mathcal{O}_X(2)) = 8$ , and the vector space in the middle has dimension 10, so we conclude that  $h^0(\mathbb{P}^3, \mathcal{I}(2)) \geq 2$ . An element of  $H^0(\mathbb{P}^3, \mathcal{I}(2))$  is a form of degree 2, whose zero-set is a quadric surface containing  $X$ . Any such quadric must be irreducible and reduced because  $X$  is not contained in any hyperplane. Hence,  $X$  is contained in two distinct irreducible quadric surfaces and is a complete intersection because  $X$  is a curve of degree 4.

□

9. Prove the following lemma of Bertini: if  $X$  is a curve of degree  $d$  in  $\mathbb{P}^3$ , not contained in any plane, then for almost all planes  $H \subseteq \mathbb{P}^3$  (meaning a Zariski open subset of the dual projective space  $(\mathbb{P}^3)^*$ ), the intersection  $X \cap H$  consists of exactly  $d$  distinct points, no three of which are collinear.

*Proof.* Bertini's theorem says a general plane in  $\mathbb{P}^3$  intersects  $X$  in  $d$  distinct points. To show that no three of these points are collinear generically, it is enough to show a general plane in  $\mathbb{P}^3$  does not contain a multisequant of  $X$ . The proof of (3.10) shows that the set of multisequants of  $X$  has dimension  $\leq 1$  in  $(\mathbb{P}^3)^*$ , and a general plane in  $\mathbb{P}^3$  corresponds to a line in  $(\mathbb{P}^3)^*$ . Hence, for almost all planes  $H \subseteq \mathbb{P}^3$ , the intersection  $X \cap H$  consists of exactly  $d$  distinct points, no three of which are collinear. □

12. For each value of  $d = 2, 3, 4, 5$  and  $r$  satisfying  $0 \leq r \leq \frac{1}{2}(d-1)(d-2)$ , show that there exists an irreducible plane curve of degree  $d$  with  $r$  nodes and no other singularities.

*Proof.* There is nothing to prove for  $d = 2$ , so fix any  $d \geq 3$ , and denote  $g = \frac{1}{2}(d-1)(d-2)$ .

$$d = 2, 3, 4, 5, \quad g = 0, 1, 3, 6$$

Both extremes  $r = 0, g$  are constructed in (3.11.1).

If  $X$  is any elliptic curve and  $P$  is any point in  $X$ , then the linear system  $|dP|$  gives an embedding of  $X$  in  $\mathbb{P}^{d-1}$  as a curve of degree  $d$ . Projecting the curve into  $\mathbb{P}^2$  by (3.5) and (3.10) gives a plane curve of degree  $d$  and  $g-1$  nodes. This gives  $d = 4$  and  $r = 2$ .

By (Ex 3.6), the complete intersection of two quadric surfaces in  $\mathbb{P}^3$  is an elliptic curve of degree 4. Projecting from a point not in the curve by (3.10) gives a plane curve of degree  $d = 4$  and  $r = 2$  nodes.

Similarly, any divisor of degree  $\geq 5$  on a genus 2 curve is very ample by (Ex. 1.5), so there exists plane curves of degree  $d \geq 5$  with  $g-2$  nodes. This gives  $d = 5, r = 4$ .

If  $Q \subset \mathbb{P}^3$  is the nonsingular quadric surface, then there exists divisors of type  $(a, b)$  for any  $a, b > 0$  that are linearly equivalent to an irreducible nonsingular curve of degree  $a+b$  and genus  $(a-1)(b-1)$  as a curve in  $\mathbb{P}^3$ .

A divisor of type  $(2, 3)$  gives a nonsingular curve of degree  $2+3=5$  and genus  $(2-1)(3-1)=2$  in  $\mathbb{P}^3$ . Projecting into  $\mathbb{P}^2$  by (3.10) gives a plane curve of degree  $d=5$  and  $r=4$  nodes. Projecting this curve from one of its own points that does not lie in a plane spanned by coplanar tangent lines gives a plane curve of degree  $d=4$  and  $r=1$  node.

Now, consider a divisor of type  $(3, 3)$ . It is a nonsingular curve of degree  $3+3=6$  and genus  $(3-1)(3-1)=4$  in  $\mathbb{P}^3$ . Projecting this curve from one of its points that does not lie in a plane spanned by coplanar tangent lines gives a plane curve of degree  $d=5$  and  $r=2$  nodes.

Finally, consider the divisor of type  $(4, 2)$ . It has degree  $2+4=6$  and genus  $(4-1)(2-1)=3$  as a curve in  $\mathbb{P}^3$ . Projecting this curve from one of its points that does not lie in a plane spanned by coplanar tangent lines gives a plane curve of degree  $d=5$  and  $r=3$  nodes.

More generally, any divisor of degree  $d \geq 2g+1$  on any curve of genus  $g$  is very ample by (3.2), so there exists plane curves of degree  $d$  and  $g-d$  nodes. This gives  $d=5$  and  $r=1$ .

□